Quiz 1 - Set 2

Graded

Student

Om Sanan

Total Points

8 / 11 pts

Question 1

1 — **1** / 1 pt

- ✔ **– 0 pts** Correct
- **– 1 pt** Incorrect

Question 2

2 — **1** / 1 pt

- ✔ **– 0 pts** Correct
- **– 1 pt** Incorrect

Question 3

3 — **1** / 1 pt

- ✔ **– 0 pts** Correct
- **– 1 pt** Incorrect

Question 4

4 — **1** / 3 pts

4.1 4i — **0** / 1 pt

- **– 0 pts** Correct
- ✔ **– 1 pt** Incorrect

4.2 4ii — **1** / 1 pt

- ✔ **– 0 pts** Correct
- **– 1 pt** Incorrect

4.3 4iii — **0** / 1 pt

- **– 0 pts** Correct
- ✔ **– 1 pt** Incorrect

Question 5

5 **1** / 2 pts

5.1 **5i** **1** / 1 pt

✔ **– 0 pts** Correct

– 1 pt Incorrect

5.2 **5ii** **0** / 1 pt

– 0 pts Correct

✔ **– 1 pt** Incorrect

Question 6

6 **1** / 1 pt

✔ **– 0 pts** Correct

– 1 pt Incorrect

Question 7

7 **1** / 1 pt

✔ **– 0 pts** Correct

– 1 pt Incorrect

Question 8

8 **1** / 1 pt

✔ **– 0 pts** Correct

– 1 pt Incorrect

-3

1/27/26

EE 148a [Winter 2026]
Quiz 1
Set: 2

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Instructions: Choose exactly one answer for each question. No collaboration.

✓1. (1 pt) A binary perceptron classifies using $\hat{y} = \text{sign}(w^\top x + b)$. The decision boundary is the set of points where the predicted class changes. What geometric object is the decision boundary?

(a) A finite set of isolated points
(b) A parabola
(c) A hyperplane defined by $w^\top x + b = 0$
(d) A circle (in any dimension)

✓2. (1 pt) Which labeling of points in $\mathbb{R}^2$ *cannot* be perfectly classified by a single-layer perceptron (with any choice of w, b)?

(a) OR gate labeling
(b) AND gate labeling
(c) Any linearly separable labeling
(d) XOR gate labeling

✓3. (1 pt) Let $L = (w_1 + w_2) \cdot w_3$. Which set of partial derivatives is correct?

(a) $\frac{\partial L}{\partial w_1} = w_2 w_3,\ \frac{\partial L}{\partial w_2} = w_1 w_3,\ \frac{\partial L}{\partial w_3} = w_1 w_2$
(b) $\frac{\partial L}{\partial w_1} = w_1 + w_2,\ \frac{\partial L}{\partial w_2} = w_1 + w_2,\ \frac{\partial L}{\partial w_3} = w_3$
(c) $\frac{\partial L}{\partial w_1} = w_1,\ \frac{\partial L}{\partial w_2} = w_2,\ \frac{\partial L}{\partial w_3} = w_3$
(d) $\frac{\partial L}{\partial w_1} = w_3,\ \frac{\partial L}{\partial w_2} = w_3,\ \frac{\partial L}{\partial w_3} = w_1 + w_2$

4. (3 pts) A K-class classifier outputs logits $z \in \mathbb{R}^K$, and probabilities

$$p_k = \frac{e^{z_k}}{\sum_{j=1}^{K} e^{z_j}} \quad (k = 1, \ldots, K).$$

For a single training example with one-hot label y (true class index t), the cross-entropy loss is

$$L(z, y) = -\sum_{k=1}^{K} y_k \log p_k.$$

Assume $0 < p_k < 1$ for all k.

✗ **(i)** (1 pt) What is the derivative of L with respect to the true-class logit z_t?

(a) $\frac{\partial L}{\partial z_t} = 1 - p_t$
(b) $\frac{\partial L}{\partial z_t} = -p_t$
(c) $\frac{\partial L}{\partial z_t} = 0$
(d) $\frac{\partial L}{\partial z_t} = p_t - 1$

✓ **(ii)** (1 pt) A gradient descent step updates $z \leftarrow z - \eta \nabla_z L$ for $\eta > 0$. What happens to z_t and a non-true logit z_k (for $k \neq t$) after one step?

(a) z_t increases and z_k decreases
(b) z_t decreases and z_k increases
(c) z_t remains unchanged and z_k decreases
(d) z_t increases and z_k remains unchanged

✗ **(iii)** (1 pt) Which modification to the logits leaves the softmax probabilities *unchanged* but improves numerical stability (i.e., avoids overflow from excessively large e^{z_k})?

(a) Replace z by $2z$
(b) Replace z by $z - \max_j z_j \mathbf{1}$
(c) Replace z by $z - \text{mean}(z) \mathbf{1}$ ←
(d) Replace z by $|z|$

5. (2 pts) You fit a linear model $\hat{y} = Xw$ to predict labels $y \in \mathbb{R}^n$ from features $X \in \mathbb{R}^{n \times d}$ (no bias term). Assume the columns of X are whitened so $X^\top X = nI$. Let $x_i \in \mathbb{R}^n$ denote the ith column of X, and define the (unnormalized) correlation score $a_i \triangleq x_i^\top y$. You consider two regularized objectives (with $\lambda > 0$):

$$J_1(w) = \|Xw - y\|_2^2 + \lambda \|w\|_1 \qquad \text{and}$$
$$J_2(w) = \|Xw - y\|_2^2 + \lambda \|w\|_2^2.$$

(i) (1 pt) Under the whitening assumption, the minimizer $w^\# = \arg\min_w J_2(w)$ has a simple closed form. Which expression is correct for its ith coordinate $w_i^\#$?

(a) $w_i^\# = \dfrac{a_i}{n - \lambda}$
(b) $w_i^\# = \dfrac{a_i}{n + \lambda}$
(c) $w_i^\# = \dfrac{a_i}{n} - \lambda$
(d) $w_i^\# = \text{sign}(a_i) \max\{|a_i| - \lambda, 0\}$

(ii) (1 pt) For $w^* = \arg\min_w J_1(w)$, which statement best describes when w_i^* becomes *exactly zero*?

Hint (you may use without proof): For any scalar $u \in \mathbb{R}$, $\alpha > 0$, $\lambda > 0$, the minimizer of

$$\min_{t \in \mathbb{R}} \ \alpha(t - u)^2 + \lambda |t|$$

is the *soft-thresholding* solution

$$t^* = \text{sign}(u) \max\left\{ |u| - \frac{\lambda}{2\alpha}, 0 \right\}.$$

(a) $w_i^* = 0$ whenever $|a_i| \le n\lambda/2$
(b) $w_i^* = 0$ whenever $|a_i| \le \lambda$
(c) $w_i^* = 0$ whenever $|a_i| \le \lambda/(2n)$
(d) $w_i^* = 0$ whenever $|a_i| \le \lambda/2$

6. (1 pt) A fully-connected network maps $\mathbb{R}^4 \to \mathbb{R}^{10} \to \mathbb{R}^3$ using affine layers (each layer has weights and biases). How many trainable parameters does it have?

(a) 73
(b) 123
(c) 83
(d) 90

7. (1 pt) Which statement best describes stochastic gradient descent (SGD) with mini-batches?

(a) It computes the exact gradient over the whole dataset every step
(b) It uses the gradient on a (random) mini-batch as a noisy estimate of the full-dataset gradient
(c) It requires second derivatives (the Hessian)
(d) It updates weights only once per epoch after seeing all data

8. (1 pt) A model outputs probabilities [0.1, 0.7, 0.2]. The true class is the *3rd* class. What is the cross-entropy loss for this example (natural log)?

(a) $-\ln(0.9) \approx 0.1$
(b) $-\ln(0.2) \approx 1.61$
(c) $-\ln(0.1) \approx 2.3$
(d) $-\ln(0.7) \approx 0.36$